

Less is more when more is more than more: A Nanosyntactic analysis of Mandarin *duōshǎo**

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Mandarin *duōshǎo* ‘how much’ is a question word/phrase with remarkable properties: the degree question marker *duō* ‘how’ is syncretic with the quantifier *duō* ‘much’, and the quantifier part *shǎo* is, elsewhere, equivalent to English *little*. Adopting a Nanosyntactic approach, we propose a homophone-free structural analysis of *duōshǎo* ‘how much’ that relates the two properties.

1 Empirical challenge

Gradable adjectives have been claimed to involve an ordered scale along a given dimension (cf. Cresswell, 1976; Kennedy, 2001). *Big* and *small*, for example, both indicate a scale of degrees along the same dimension of size, but with opposite orderings. Both members of the antonym pair can be modified by degree adverbs such as *very*. However, when combined with degree-*how*, only the positive member is felicitous in a neutral context, e.g., in case of absolute ignorance about the size of a box, it is felicitous to ask *how big is it*, but not *#how small is it* (cf. Seuren, 1978; Rett, 2007).

- (1) a. *very big / very small*
b. *how big / #how small* (neutral reading)

The same applies to gradable quantifiers such as *much* and *little*:

- (2) a. *very much/little*
b. *how much / #how little* (neutral reading)

For gradable adjectives, the generalization also holds in Mandarin. For example, *dà/xiǎo* ‘big/small’ can be modified by degree adverbs such as *hěn* ‘very’ and

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duō ‘how’, but only *dà* ‘big’ can be used in an unbiased/neutral question about the size of something.

- (3) a. *hěn dà / hěn xiǎo* ‘very big / very small’
 b. *duō dà / #duō xiǎo* (neutral reading)

However, Mandarin quantifiers show an interesting twist. While *duō* ‘much’¹ and *shǎo* ‘little’² can indeed both combine with *hěn* ‘very’ as well as with *duō* ‘how’, it is the negative member *shǎo* ‘little’ that features in the unbiased/neutral question.³

- (4) a. *hěn duō / hěn shǎo* ‘very much / very little’
 b. *#duō duō / duō shǎo* ‘how much’ (neutral reading)

The Mandarin duo *duō / shǎo* thus exhibits two puzzles :

- *duō* is ambiguous between a quantity reading ‘much’ and a degree-*how* reading
- *shǎo* ‘little’, the negative member of the antonym pair, features in the neutral/unbiased degree question, violating the positive-unmarked generalization

We will, in this paper, develop a short and sweet nanosyntactic analysis in which the first puzzle is solved without the postulate of homophonous lexical items, and the second puzzle follows as an epiphenomenal effect of our solution to the first puzzle.

2 Nanosyntax in a nutshell

Nanosyntax (Starke, 2002, 2009, 2014, 2018; Caha, 2009; De Clercq et al., 2025) is basically a framework for the structural analysis of morphosyntactic and morphosemantic observations in natural language. It combines a bottom-up derivational syntax (Chomsky, 1995; Kayne, 1994, 1998; Koopman, 2000; Starke, 2001), a cartographic articulation of phrase structure (Rizzi, 1997; Cinque, 1999; Cardinaletti and Starke, 1999; Koopman, 2000; Kayne, 2005), a sublexically compositional semantics, a.k.a. prelexical syntax (cf. Katz and Postal, 1964;

¹It is plausible that *duō* really corresponds most closely to *m-* (clearly morphemic) in *many/much*, where *-any/-uch* are related to the count-mass distinction, given that Mandarin *duō* ‘many/much’ and *shǎo* ‘few/little’ cover both the count and the mass readings. We will, however, gloss over this aspect here for expository purposes (cf. De Clercq, 2017, for a typological study of quantity words regarding the mass/count distinction).

²One might note that Mandarin *shǎo* ‘little (in quantity)’ shares [voiceless; sibilant; fricative]ǎo with *xiǎo* ‘little (in size)’. The similarity is reflected not only in the pronunciation, but also in the hanzi characters 少 and 小, and the syncretism shown in their English translation as *little*, suggesting that Mandarin *shǎo* and *xiǎo* share a morpheme.

³One may wonder if this unusual deviation may be a haplological effect (cf. Neeleman and van de Koot’s (2017) repeated morph constraint), i.e., that the degree *duō* ‘how’ and the quantifier *duō* ‘much’ cannot occur adjacent to each other, due to their identical form. Note, however, that this co-occurrence does exist in Mandarin, albeit restricted to a biased reading like ‘(just) how much’ (as indicated by the # instead of * in (4-b)). Similarly, a biased reading is also accessible to *duō shǎo/dà/xiǎo* as well as their English counterparts.

McCawley, 1968; Lakoff, 1976), and a piece-based realizational morphology (cf. Chomsky, 1957; Halle and Marantz, 1993).

The syntax is a combinatory engine that organizes atomic elements from a universal set of semantico-syntactic features into binary branching hierarchical structures (referred to as *S-trees*).

After each merger operation, the resulting S-tree is subjected to lexicalization, a procedure during which the S-tree is checked against a list of realizability templates (referred to as *L-trees*), each of which is discretely associated with phonological and, in some cases, conceptual-semantic information. These complex units of association between kinds of information relevant to different modules are referred to as *morphemes*, and are represented as exemplified in (5).

- (5) Format of morphemes:
 $\langle /bla/, \begin{array}{c} F_3 \\ \swarrow \quad \searrow \\ F_3 \quad F_2 \\ \quad \swarrow \quad \searrow \\ \quad F_2 \quad F_1 \end{array}, BLA' \rangle$

The conceptual information BLA' of morphemes, and the syntactically composed feature structure (S-tree) are the only components relevant to the semantics. I.e. crucially, the L-tree is irrelevant to the semantics.

The checking for lexicalizability consists of verifying the availability of an *L-tree* (i.e. a morpheme) that contains a structure fully isomorphic with the *S-tree* to be lexicalized. This condition is referred to as the *Superset Principle*, and formulated in (6).

- (6) Superset Principle (adapted from Starke, 2009):
 A morpheme can lexicalize a syntactic constituent iff its L-tree contains said constituent (i.e. the S-tree).

Lexicalizable structures are lexicalized by one morpheme with a compatible *L-tree*, and the syntactic derivation continues by merger of the next relevant feature in accordance with a universal functional sequence (referred to as *fseq*). If multiple compatible morphemes are available for the lexicalization of a given S-tree, the best match is chosen in accordance with a *Simplicity Preference Principle*, SPP (7). We are formulating the SPP in a lexicalization specific fashion. However, it clearly belongs to the Paninian family of grammatical principles.⁴ We will not give a formally precise algorithm for calculating simplicity, but simply assume, for present purposes, that the fewer features a structure contains,

⁴Our SPP proposal is very close in spirit to Cardinaletti and Starke's (1994) *Minimize α* , Starke's (2009) *Minimize Junk*, all being functional adaptations of the Elsewhere Principle (cf. Kiparsky, 1973): *In case two rules, R1 and R2, can apply in an environment E, R1 takes precedence over R2 if it applies in a proper subset of environments compared to R2* (formulation taken from De Clercq et al. (2025)). The actual Elsewhere Principle, however, would not work for our purposes, since the relevant competitors are not fully in a superset-subset relation (cf. (11) and (14) in Section 3.2).

the simpler it is, and that a complex left branch counts as more complex than a simple left branch.

- (7) Simplicity Preference Principle:
In case two morphemes can lexicalize an S-tree, the one with the simpler L-tree takes precedence over the one with the more complex L-tree.

Non-lexicalizability of an S-tree triggers a sequence of syntactic repair options, as specified in the *Lexicalization Algorithm* (Starke, 2018; De Clercq et al., 2025), of which we are adopting a version similar to that proposed in Caha et al. (2019).

- (8) Lexicalization Algorithm:
- a. Merge F and lexicalize.
 - b. If fail, try a spec-to-spec movement and lexicalize.
 - c. If fail, try a movement of the complement of the newly inserted feature and lexicalize.
 - d. If fail, go back to the previous cycle, and try the next option for that cycle.
 - e. If fail, spawn a new derivation starting with the last successfully lexicalized feature, F^{-1} , merge F, and then merge FP back with the initial derivation, projecting feature F to the top node.

Importantly, while there is debate over various ideas on how a new derivation (in a first or in a second workspace) starts, we are reformulating (8-e) to reflect the proposal in Caha et al. (2019) that the new derivation must start with the last successfully lexicalized feature from the original derivation.⁵

3 From *hěn duō* ‘very much’ to *duō shǎo* ‘how much’

The intuition behind our analysis of *duō* ‘much’ and *shǎo* ‘little’ (as adjectival quantifiers) is that both denote a departure from some contextual (perhaps lexico-pragmatic) default quantity. While *duō* ‘much’ denotes a departure in a positive direction, *shǎo* ‘little’ denotes a departure in a negative direction.

We are, of course, aware that our clumsy “departure from” and “direction” are stand-ins for complex and partly unresolved debates in compositional semantics (cf. Wellwood, 2020). What is relevant for us is that both *duō* and *shǎo* share a common semantic core, i.e. a level/degree of quantity. This semantic

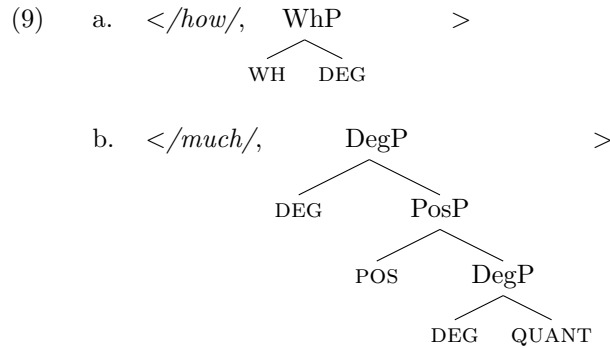
⁵This, technically, makes the two derivations featurally overlap (albeit only minimally) in a rather specific way. Given that the S-tree in derivation 2 ends up as a specifier of the S-tree in derivation 1, this is curiously reminiscent of a Spec-Head configuration, which played a major role in the 1980s/90s as one of the two local structural configurations in grammar (cf. Koopman, 2006): Head-Complement for *selection*; Specifier-Head for *identity* (ϕ -agreement (Kayne, 1989); WH-criterion (Rizzi, 1996); NEG-criterion (Haegeman and Zanuttini, 1991); quantifier scope (Beghelli and Stowell, 1997)).

core is, we propose, what is projected in the compositional semantics of the neutral question *how much*. No positive or negative departure from a contextual default quantity is part of the denotation of the neutral *how much*.

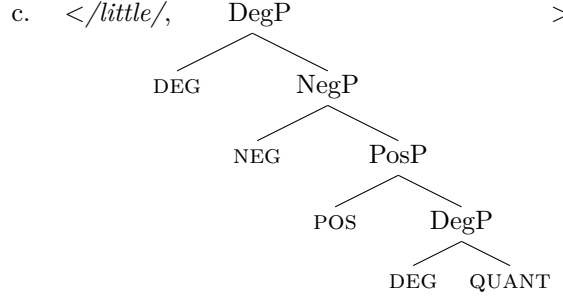
3.1 English *how much*

English has *many/much* where Mandarin has *duō*, and *few/little* where Mandarin has *shǎo*. (We set aside the mass vs count question, cf. footnote 1.) An initial observation we need to get clarity on before continuing is that of why English asks *how much* rather than *how little* as a non-biased context-neutral inquiry for the degree of quantity. This, of course, seems to be but a special case of the more general pattern that, in antonym pairs of gradable adjectives, it is the “positive” member that also allows the unmarked reading (cf. Bolinger, 1977; Lehrer, 1985; Kennedy, 2001): *how old*; *how tall*; *how fast*, etc. Many different ways in which one might account for this are thinkable (and have been thought and proposed). It could be a matter of pragmatics or it could be a matter of lexical semantics. It could or could not be structurally represented in the syntax. Importantly for us: this is not English-specific; it is, in principle, the same in Mandarin (and possibly universal).

We will focus exclusively on *much/duō*, and *little/shǎo*, and adopt (a variant of) the idea that *little* is *not much* (cf. *few* as *not many* Barwise and Cooper, 1981; Solt, 2006; Caha et al., 2019), where both *little* and *much* are degree adjectives (cf. Kayne, 2007), i.e. they are DegPs. Nanosyntactically this is captured by the L-trees in (9).⁶ Recall that the L-trees are not semantically interpreted. The semantic interpretation in Nanosyntax is a principled combination of the compositional semantics represented by the S-tree and the conceptual semantics from morphemes that are specified as such.



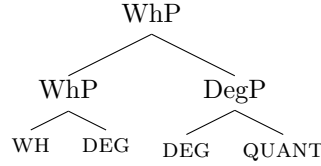
⁶The careful reader will notice that Deg is recursive here. The rationale for this proposal will be explicated in Section 3.2.



The representation of *little* as *much* with an additional [NEG], now, provides the means for modelling the fact that English (and in principle any language that has the relevant words) uses *how much* rather than *how little* for the neutral (i.e. non-biased) reading.

We will assume (as we naturally will in our analysis of Mandarin too) that the syntactic derivation of the neutral reading of *how much* only combines the features [QUANT], [DEG] and [WH], and that a bi-morphemic realization thereof reflects the fact that the language in question does not have a single morpheme that can realize the structure, i.e. that the procedure (b)-(e) of the Lexicalization Algorithm (8) applies, eventually deriving (10).

(10) S-tree for *how much*



In (10), the right branch DegP only contains the “degree of quantity” core, which is contained in both L-trees (9-b) and (9-c). But since (9-c) has an additional [NEG] as compared to (9-b), the competition for lexicalization of the right branch in (10) will be naturally resolved in favour of (9-b) by the Simplicity Preference Principle (7).

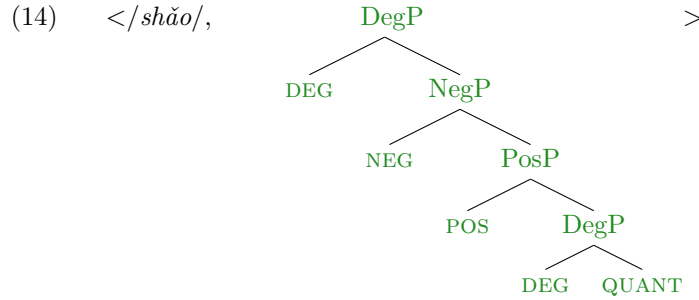
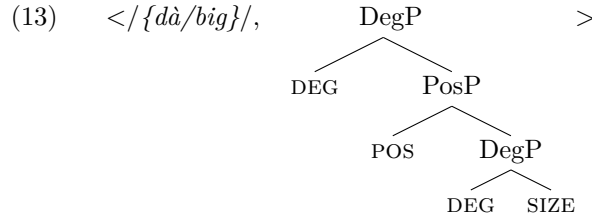
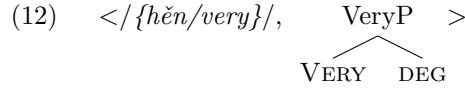
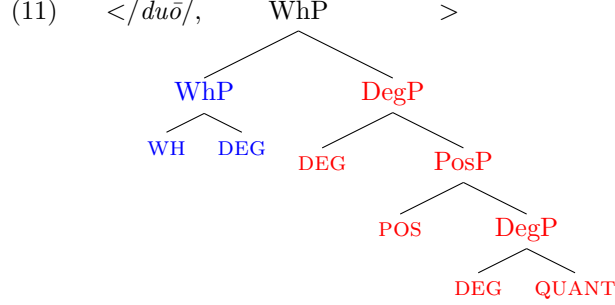
With this as an analytical background, we can now tackle the question of why Mandarin seems to say *how little* when meaning *how much* (in the neutral reading).

3.2 Lexical entries for *duō* and *shǎo*

We are proposing the following lexical entries for *duō* and *shǎo*. Lexical entries are, of course, necessary stipulations. But they have a specific format and are subject to certain constraints within which the creative freedom of the analysts searches for the version which, in combination with the empirical imperative and their assumptions about other blindly operating modules, derives the correct forms.

There are a few features in our proposal that we want to draw the reader’s attention to: (i) the complex left branch in (11), which models the ambiguity of

duō as syncretic; (ii) the repetition of [DEG], accounting for the fact that both the neutral and the biased reading are syntacticosemantically degree phrases; and (iii) the partial similarity between (11) and (14), cf. (9-b) and (9-c).



Note that the L-trees (12) and (13) are identical for Mandarin and English. This is because the comparative contrast we are interested in in this paper concerns (11) and (14).⁷ Now that we are ready to build structures and lexicalize them, let's first derive (15-a), then (15-b), and eventually venture on the derivation of (15-c).

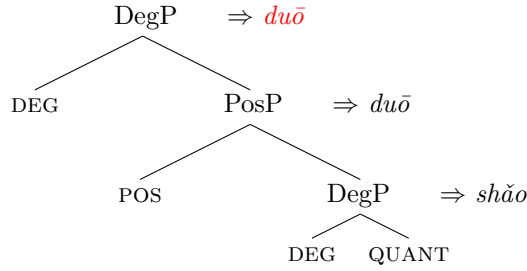
⁷The identity of Mandarin and English with regard to (12) and (13) is most certainly a simplification, but irrelevant to the present discussion.

- (15) a. *hěn duō* ‘very much’
 b. *duō dà* ‘how big’
 c. *duō shǎo* ‘how much’

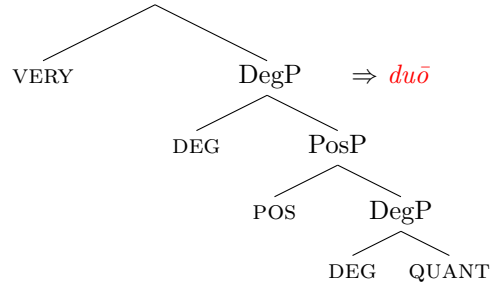
3.3 Deriving *hěn duō* ‘very much’

Let us first derive (15-a): *hěn duō* ‘very much’

- (16) a. First the syntax incrementally builds a positively ordered DegP, with a QUANT bottom, which is sent for lexicalization after each merger operation (i.e. after merger of the lower [DEG], of [POS], and of the higher [DEG]). The lower DegP is contained in two L-trees, (11) and (14). At that stage, (14) wins, because it is simpler, compared to (11), which has a complex left branch, cf. SPP in (7). [This is crucial! It is the basis for the right analysis of *duōshǎo*.] However, this first lexicalization remains opaque in the derivation of (15-a), because once [POS] is merged, only (11) can lexicalize it, and *duō* ‘much’ is interpreted as a positively ordered quantifier.



- b. Then the syntax merges *very*:⁸ veryP → This can't lexicalize.



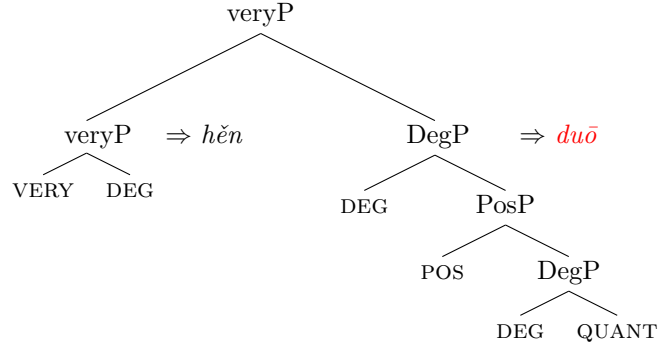
- c. The Lexicalization Algorithm (8) eventually⁹ undoes merge-*very*,

⁸We're setting aside semantic and syntactic questions regarding the *veryP*, which is just a shorthand here.

⁹See (8). For the sake of simplicity, we are skipping over the steps that remain opaque in the present derivations, in which the procedure needs to go all the way to (8-e), deriving a complex left branch.

and opens a new derivation, starting with the last successfully lexicalized feature, [DEG].

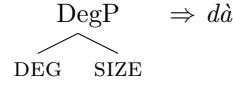
After merger of *very*, the structure in derivation 2 lexicalizes as *hěn* by (12), and is merged back with the structure in derivation 1, projecting *veryP*.



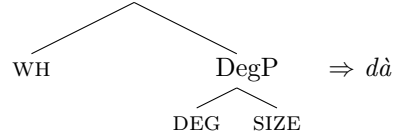
3.4 Deriving *duō dà* ‘how big’

Let us next derive (15-b): *duō dà* ‘how big’

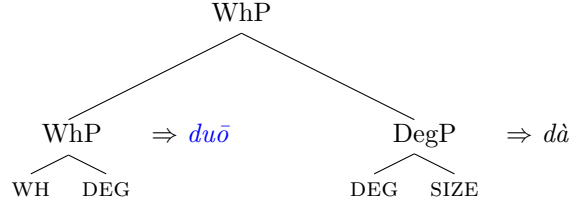
- (17) a. First the syntax builds the lower/neutral *DegP*, with a [SIZE] bottom, which is sent for lexicalization.



- b. Then the syntax adds [WH]: *WhP* → This can’t lexicalize.



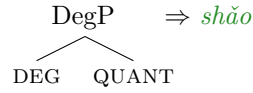
- c. The Lexicalization Algorithm (eventually) undoes merge-[WH], and opens a second derivation, starting with the last successfully lexicalized feature, [DEG].
The structure built in derivation 2 lexicalizes as *duō* by (11), and is merged back with the structure in derivation 1, projecting *WhP*.



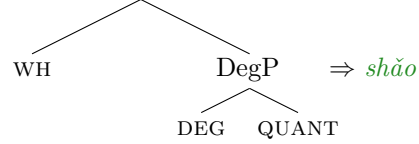
3.5 Deriving *duō shǎo* ‘how much’

Let us now attempt to derive (15-c): *duō shǎo* ‘how much’

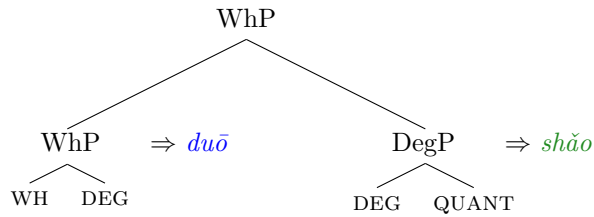
- (18) a. First the syntax builds the lower/neutral DegP, with a [QUANT] bottom, which is sent for lexicalization.



- b. Then the syntax adds [WH]: **WhP** → This can’t lexicalize.



- c. The Lexicalization Algorithm (eventually) undoes merge-[WH], and opens a second derivation, starting with the last successfully lexicalized feature, [DEG].
The structure built in derivation 2 lexicalizes as *duō* by (11), and is merged back with the structure in derivation 1, projecting WhP.



This analysis is transparently compositional, homophone-free, and, importantly, derives the unexpected appearance of *shǎo* ‘little’ instead of the expected *duō* ‘much’ in *duōshǎo* ‘how much’, due to the fact that *duō* itself is ambiguous between the quantifier ‘many’ and degree-*how*.

3.6 Prediction

The L-tree of Mandarin *duō* has a complex left branch by means of which its ambiguity between an amount quantifier and a degree-*wh* operator is encoded. For this reason it is more complex than the L-tree for English *much*. For the same reason the L-tree of *duō* is even more complex than that of *shǎo*, whereas the L-tree of English *much* is simpler than that of *little*.

By the Simplicity Preference Principle (7), thus, the primacy of *shǎo* (14) for non-biased DegP, and therefore the surface-real syncretism of *shǎo* is analytically linked to the ambiguity of *duō*.

This analysis of *duōshǎo*, hence, creates the typological expectation that the observed ambiguity of *duō* (which we may call an **intersective syncretism**) and the ambiguity of *shǎo* (which we may call an **containment-based syncretism**) are structurally related: *shǎo* (14) wins for lexicalizing the lower/neutral DegP because *duō* (11) has a complex left branch.

4 Conclusion

We have taken a close look at Mandarin *duōshǎo* ‘how much’ and noted two ambiguity-related puzzles. One is that *duō* can be either an amount quantifier ‘much’ or a degree-*wh* operator ‘degree-how’. The other is that in unbiased/neutral amount questions, Mandarin uses the negative member of the opposite pair instead of the positive one, which contrasts with patterns observed in other languages and even within other ontological domains also in Mandarin.

To address these puzzles, we proposed a Nanosyntactic analysis that derives the correct forms in a transparently compositional way. Interestingly, our findings reveal that these puzzles are analytically related; the second emerges as an epiphenomenon stemming from our solution to the first.

We might conclude that 有时少即是多 *yǒushí shǎo jí shì duō* ‘sometimes less is more’. Crucially, less is more only when more is more than more.

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